

Roll No.

TME-702/TAE-703

**B. TECH. (ME/AME)
(SEVENTH SEMESTER)**

MID SEMESTER EXAMINATION, 2019

CAD/CAM

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain the design phase of product life cycle.

(b) What are the applications of computers in design ?

(c) Explain the CAM phase of product life cycle.

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What are the different CAD tools through which design process is carried out ?
 - (b) What are the benefits of CAD ?
 - (c) Explain the GKS and IGES graphic standard.
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Write the differences between wireframe and solid modeling.
 - (b) Develop the equation for Bezier curve with the six control points.
 - (c) What are the different types of input devices used for giving text and graphics information to computer ? Explain any *two* types.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Find the point on the ruled surface at $u = 0.5$ and $v = 0.6$ whose boundaries are given by two Bezier curves. The first Bezier curve is defined by four control points (6, 30), (9, 20), (25, 21) and

(3)

(30, 31). The second Bezier curve is given by 3 control points (5, 5), (10, 9) and (30, 5).

(b) Write the equation for Bezier curve. Also write down any *three* characteristics of Bezier curve.

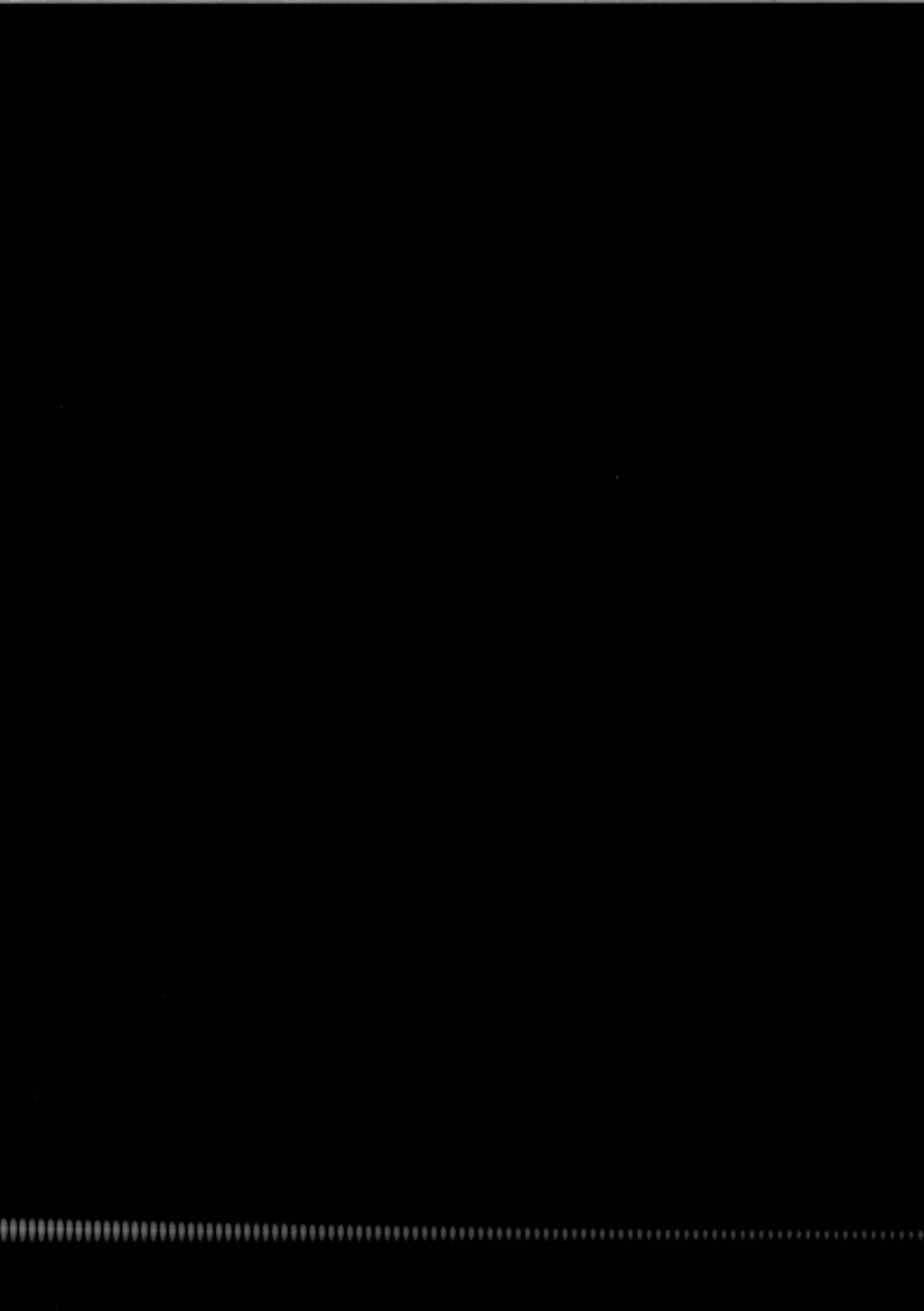
(c) What are the different function of a graphic package ?

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Write the equation for B-spline curve. Also write down any *three* characteristics of B-spline curve.

(b) Draw the schematic view of cathode ray tube (CRT). What are the different types of CRT screen scans ?

(c) A square having end points A (1, 1), B (6, 1), C (6, 6), D (1, 6) is rotated by 50° in clockwise direction keeping point (6, 1) fixed. Find the final coordinates of A.



Roll No.

TME-711

B. Tech. (ME) (SEVENTH SEMESTER) MID SEMESTER EXAMINATION, 2019

ADVANCE WELDING TECHNOLOGY

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Define welding. Classify welding based on source of energy used.

(b) What are the different joining processes ? Compare them with welding as joining process.

(c) Draw schematic diagram of oxy-acetylene gas welding process and explain in brief the working process.

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Draw schematic diagram of SMAW Process and explain its working in brief.
 - (b) Describe submerged arc welding process with neat sketch.
 - (c) Compare submerged arc welding and shielded metal arc welding process.
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Describe the working of TIG welding process with neat sketch.
 - (b) Describe the working of MIG welding process with neat sketch.
 - (c) Name ray energy welding processes. Explain any *one* of them.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain the working principle of resistance welding process with neat sketch.
 - (b) Describe explosion welding process and its limitation,
 - (c) Differentiate between SOLDERING and BRAZING.

(3)

5. - Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What are the principal functions of the coating of coated electrode ?
- (b) Describe electro slag welding process and its limitation.
- (c) Describe under water welding.

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TME-712

**B. Tech. (ME) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
COMPUTATIONAL FLUID DYNAMICS**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any two subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are ten.

1. Attempt any two parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Discuss the importance of Computational Fluid Dynamics in modeling and simulation in various fluid flow and heat transfer processes.

(b) Discuss the advantages and disadvantages of Prandtl's mixing length model.

(c) Derive the zero equation turbulence model.

(2)

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2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Discuss the grid generation methodology for the flow at the vicinity of the wall and far from wall.
 - (b) Discuss the necessary and sufficient condition for the separation of boundary layer for the flow over curved surface with adverse pressure gradient.
 - (c) What are the limitations of Finite Difference Method used in Computational Fluid Dynamics ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Discuss the Reynolds Averaging techniques and apply on Navier-Stokes equation.
 - (b) Discuss the different discretization schemes in detail.
 - (c) Describe the different types of boundary conditions used in CFD.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What are the differences between Finite Volume method and Finite Element method ?

(3)

- (b) Discuss the Reynolds stresses in detail which results after the Reynolds Averaging of Navier-Stokes equation.
 - (c) Write 3-dimensional Navier-Stokes equation and describe the importance of each term.
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Discuss and deduce the LAW of the WALL in detail considering the flow over flat surface.
 - (b) Derive the Forward Difference Scheme.
 - (c) What are the *two* equation turbulence models used in CFD ?

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**B. Tech. (ME) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
COMPUTATIONAL FLUID DYNAMICS**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

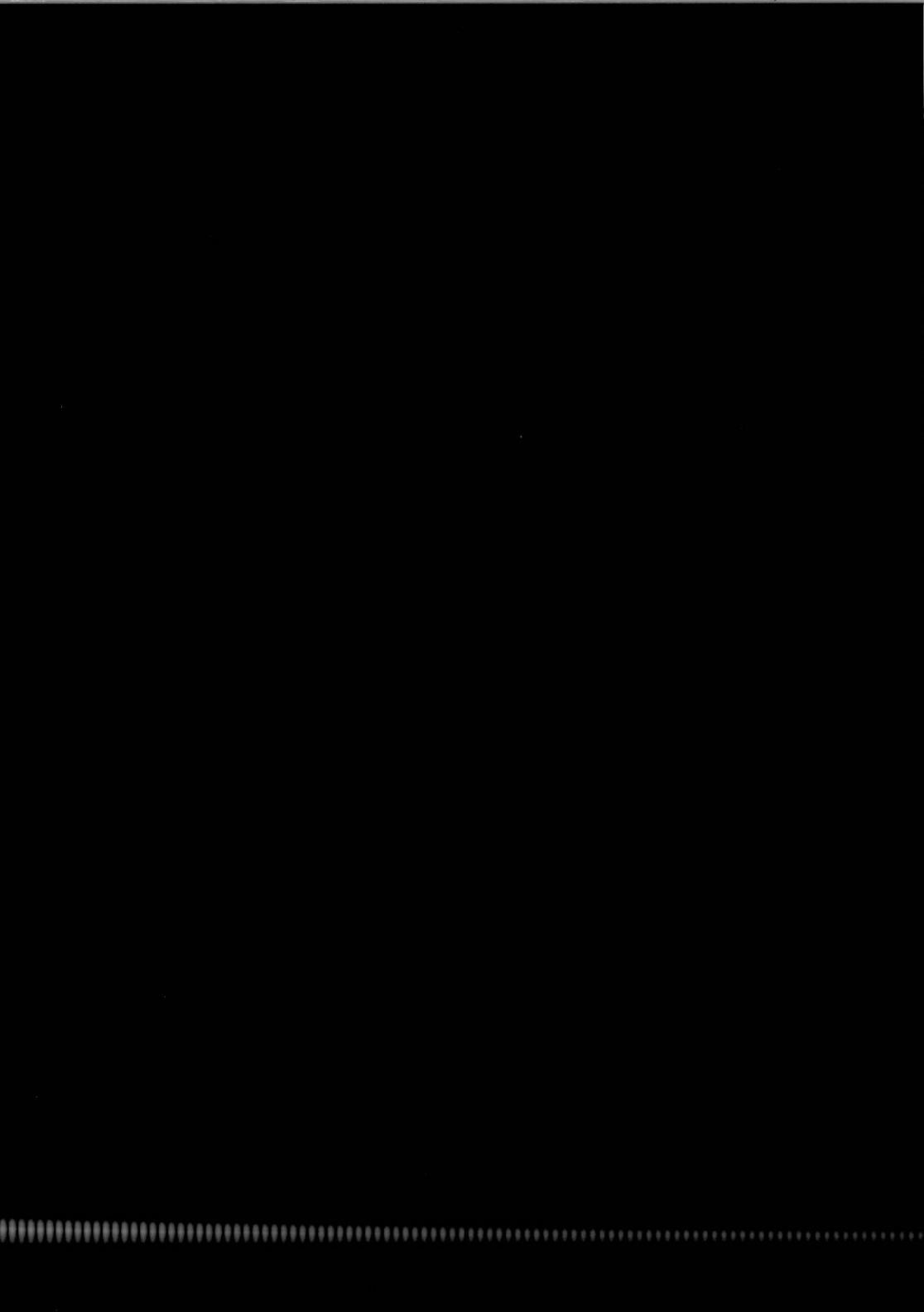
(iii) Total marks for each main question are ten.

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 - (a) Discuss the importance of Computational Fluid Dynamics in modeling and simulation in various fluid flow and heat transfer processes.
 - (b) Discuss the advantages and disadvantages of Prandtl's mixing length model.
 - (c) Derive the zero equation turbulence model.

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Roll No.

TAE-711

**B. TECH. (AME)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
QUALITY ENGINEERING AND
MANAGEMENT**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

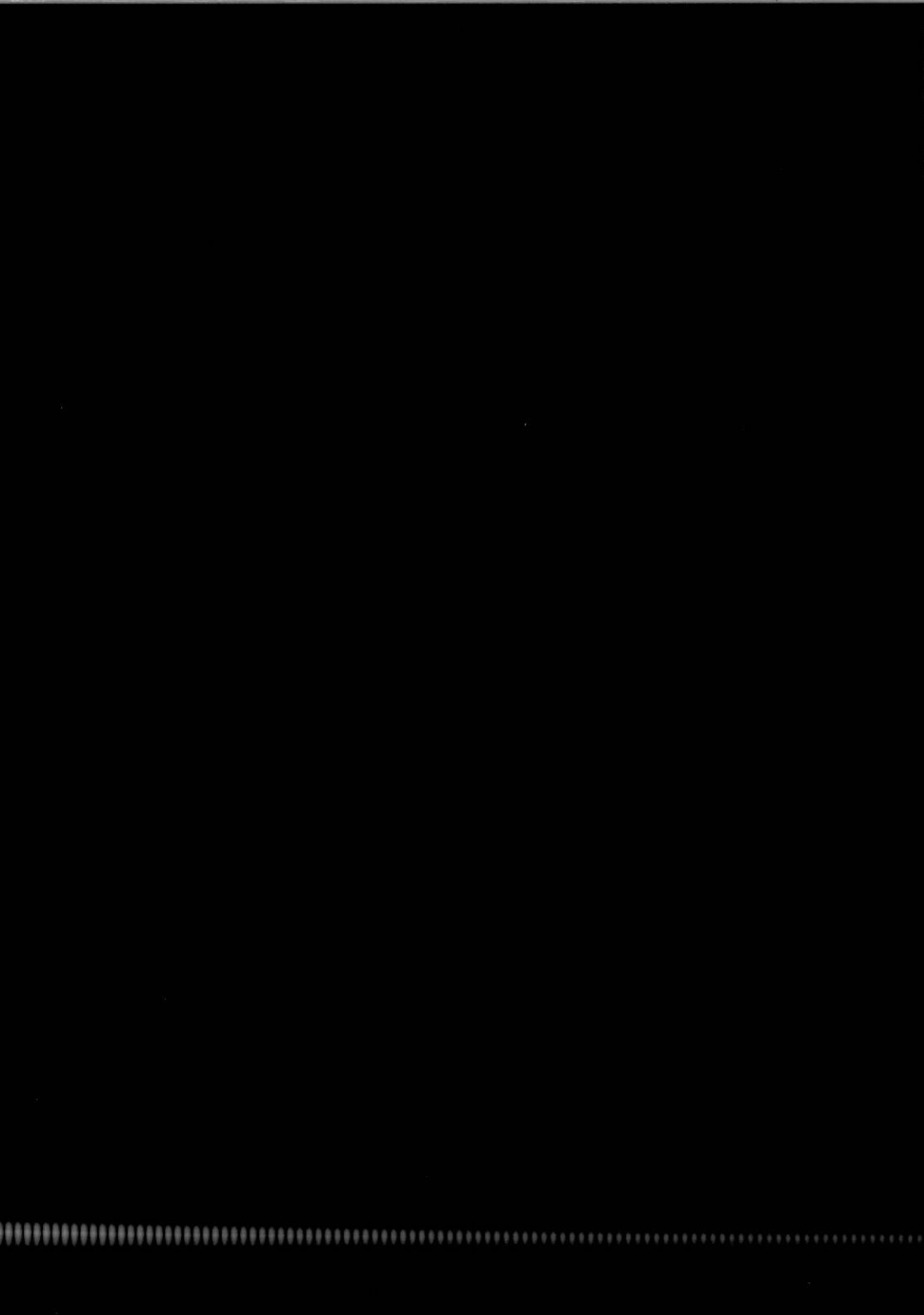
1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
 - (a) Define quality and quality management.
 - (b) Draw conceptual framework for quality management.
 - (c) Write a short note on quality improvement practices in Indian industries.

(2)

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Define Strategic Quality Management (SQM).
 - (b) What are the benefits of benchmarking ?
 - (c) Define the Quality System Standards. How does it help to improve the quality of products ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Define and give the relation between Quality and Reliability.
 - (b) Draw the Quality Action Cycle (QAC) or cycle of improvement.
 - (c) Write a short note on TQM.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Define and give difference between Quality Assurance and Quality Control.
 - (b) Write a short note on quality audits.
 - (c) Write a short note on ISO 14000.
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Write the contribution of Juran for TQM.
 - (b) List seven tools for quality management.
 - (c) Explain "Quality is Free".

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9:30 to 12:30 PM

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TAE-704

B. TECH. (AME)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, 2019

HYDRAULIC AND PNEUMATIC SYSTEMS

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain the Newton's law of viscosity. Also define term dynamic viscosity.

(b) Calculate the capillary effect in mm in a glass tube of 3 mm diameter when immersed vertically in (i) water and (b) mercury. Take surface tension of mercury and water as 0.725 N/m and

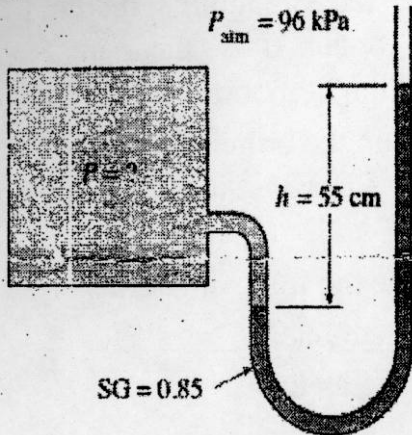
0.52 N/m respectively in contact with air. Specific gravity of mercury is given as 13.6.

- (c) Explain the phenomenon of capillarity. Obtain an expression for capillary rise/fall of a liquid.
2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain the effect of temperature of viscosity of water and air.
- (b) State and prove the Pascal's law of hydrostatic.
- (c) Explain the following :
- (i) Gauge pressure
 - (ii) Absolute pressure
 - (iii) Vacuum pressure
- with the help of a sketch. Also write the relationship among them.
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain different types of pressure measuring devices.
- (b) A manometer is used to measure the pressure of a gas in a tank. The fluid used has a specific gravity of 0.85 and the manometer column height is 55 cm, as shown in Fig. If the local atmospheric

(3)

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pressure is 96 kPa, determine the absolute pressure (P) within the tank.



(c) Explain the different types of fluid flow.

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Define the equation of continuity. Discuss its 1-dimensional form.

(b) The diameter of a pipe at the sections 1 and 2 are 10 cm and 15 cm respectively. Find the discharge through the pipe if the velocity of the water flowing through the pipe at section 1 is 5 m/s. Determine also the velocity at section 2.

(c) Define the following terms :

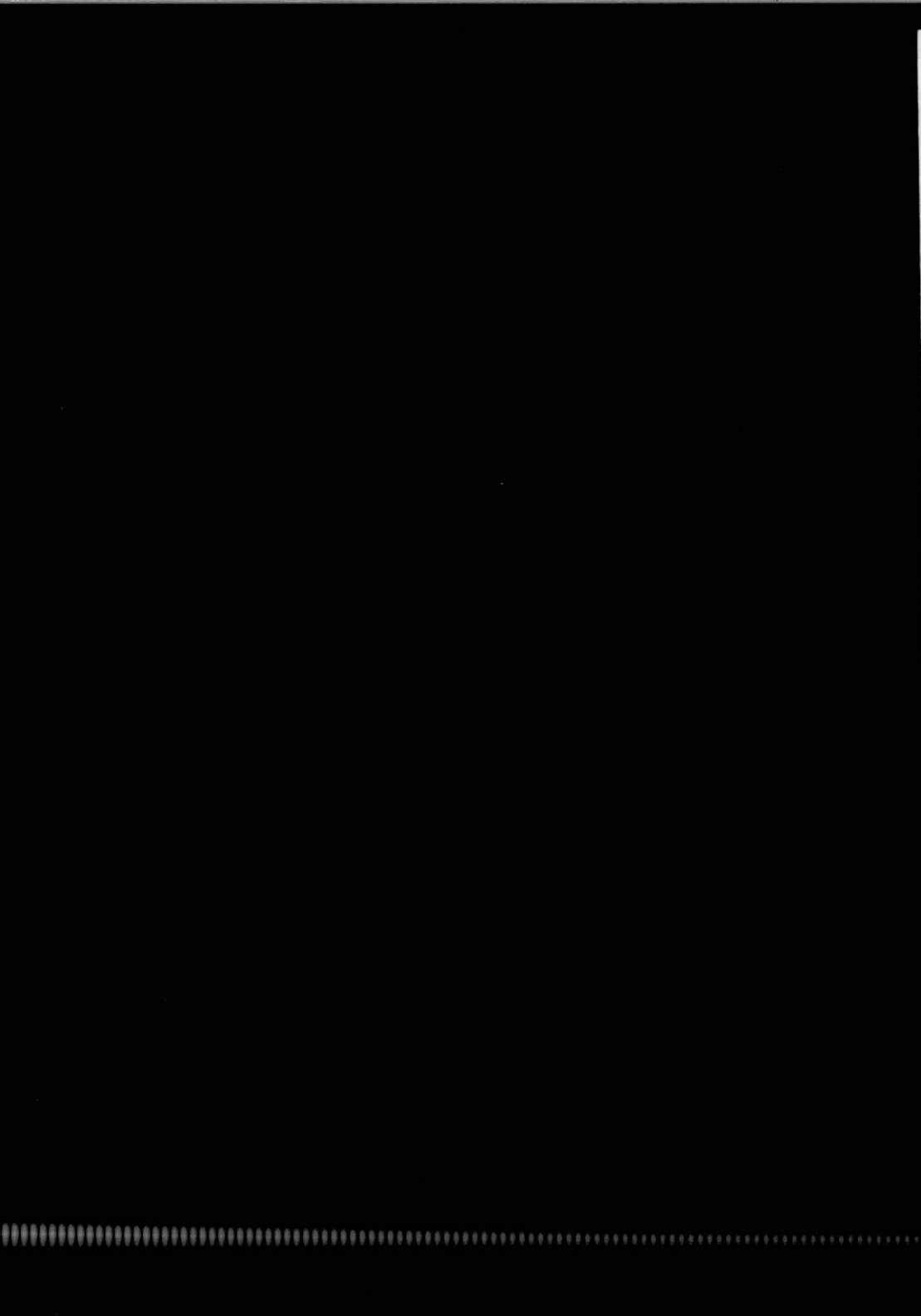
(i) Velocity Potential Function

(ii) Stream Function



5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) State the Bernoulli's theorem. How will you obtain this theorem from Euler's equation of motion ?
- (b) Explain the principle of venturimeter. Also write an expression for discharge through it.
- (c) Define the following terms :
 - (i) Path line
 - (ii) Streak line
 - (iii) Stream line



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TAE-701

B. Tech. (AME) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, 2019 VEHICLE TRANSPORT MANAGEMENT

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Describe the forms of ownership with suitable examples.

(b) Write down the principle of transport management. Also explain the purpose of transport management.

(c) What are the responsibilities of an organization toward its employees ?

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What are the key points which you will keep in mind; during recruitment and training new employees ? Take example of driver and mechanic hiring.

(b) Explain the basic principles of supervising taking suitable examples.

(c) Write down the advantages and disadvantages of single ownership.

3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Write short notes on the following :

(i) Trip leasing

(ii) Driver check list

(iii) Types of vehicle operations.

(b) Formulate planning of scheduled and unscheduled maintenance. Also explain limitations of your planning.

(c) Why work scheduling is required ? Give suitable example to justify your answer.

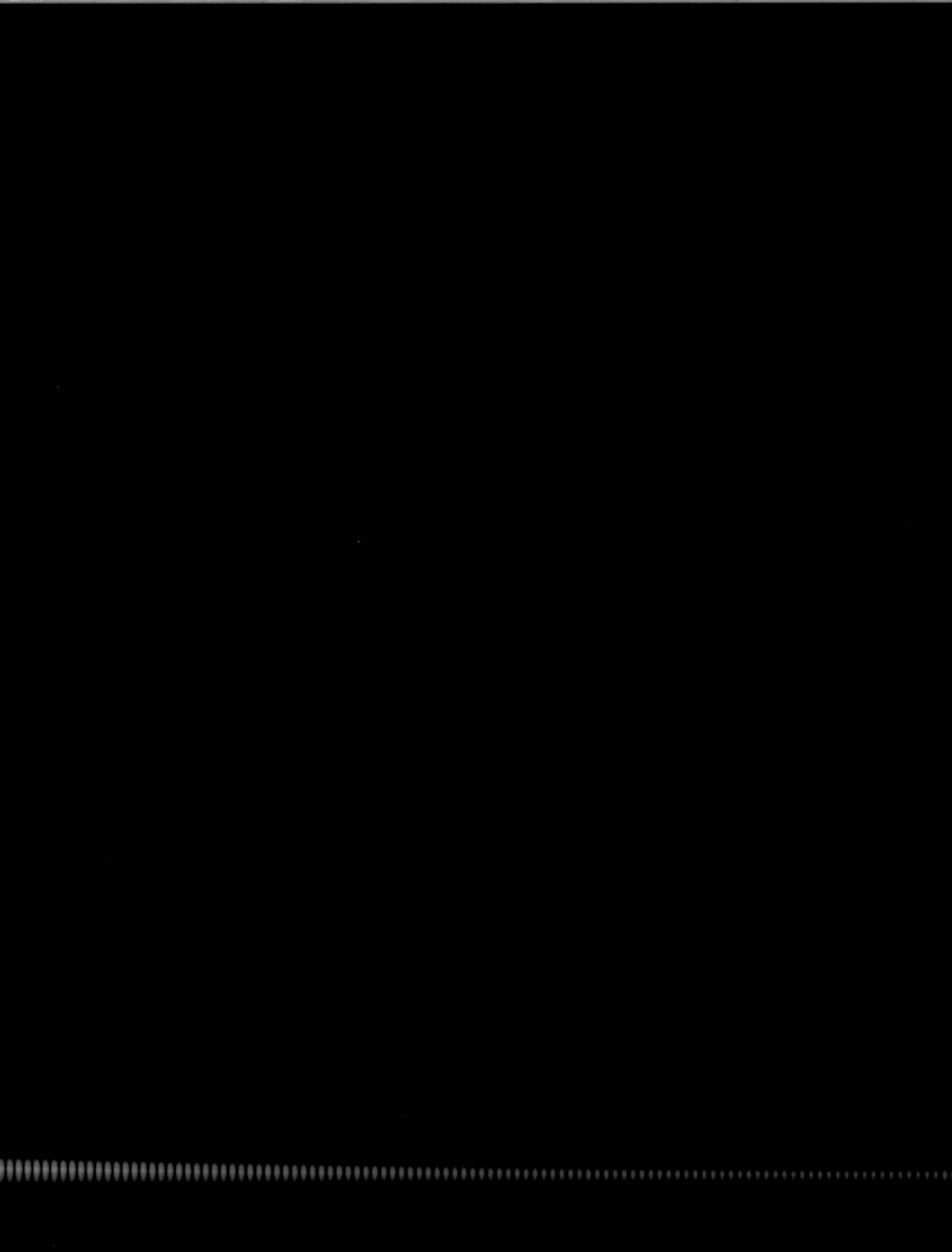
(3)

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) How you will reduce operational cost of an automobile service station ? Propose some suitable operational cost reduction model.
- (b) Design a transportation system for food products having low shelf life.
- (c) How you will do breakdown analysis ? Explain with a case study.

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) Give a detailed classification of vehicle expenses.
- (b) Explain parts control system of an automotive service station.
- (c) What are the merits and demerits of public sector companies ?



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TME-702/TAE-703

**B. TECH. (ME/AME)
(SEVENTH SEMESTER)**

MID SEMESTER EXAMINATION, 2019

CAD/CAM

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain the design phase of product life cycle.

(b) What are the applications of computers in design ?

(c) Explain the CAM phase of product life cycle.

Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What are the different CAD tools through which design process is carried out ?

(b) What are the benefits of CAD ?

(c) Explain the GKS and IGES graphic standard.

3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Write the differences between wireframe and solid modeling.

(b) Develop the equation for Bezier curve with the six control points.

(c) What are the different types of input devices used for giving text and graphics information to computer ? Explain any *two* types.

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Find the point on the ruled surface at $u = 0.5$ and $v = 0.6$ whose boundaries are given by two Bezier curves. The first Bezier curve is defined by four control points (6, 30), (9, 20), (25, 21) and

(3)

(30, 31). The second Bezier curve is given by 3 control points (5, 5), (10, 9) and (30, 5).

- (b) Write the equation for Bezier curve. Also write down any *three* characteristics of Bezier curve.
 - (c) What are the different function of a graphic package ?
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Write the equation for B-spline curve. Also write down any *three* characteristics of B-spline curve.
 - (b) Draw the schematic view of cathode ray tube (CRT). What are the different types of CRT screen scans ?
 - (c) A square having end points A (1, 1), B (6, 1), C (6, 6), D (1, 6) is rotated by 50° in clockwise direction keeping point (6, 1) fixed. Find the final coordinates of A.

Roll No.

TME-704

B. TECH. (ME) (SEVENTH SEMESTER) MID SEMESTER EXAMINATION, 2019

OPERATIONS RESEARCH AND
OPTIMIZATION TECHNIQUES

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Define Operations Research. Also given some applications of OR.

(b) A company manufactures two products X1 and X2 on three machines A, B and C. X1 require 1 hour on machine A and 1 hour on machine B and yields a revenue of ₹ 3. Product X3 requires 2 hours on machine A

and 1 hour on machine B and 1 hour on machine C and yields revenue of ₹ 5. In the coming planning period the available time of three machines A, B and C are 2000 hours, 1500 hours and 600 hours respectively. Find the optimal product mix.

- (c) Solve graphically the following linear programming problem :

Minimise :

$$z = 3x_1 + 5x_2$$

Subject to :

$$-3x_1 + 4x_2 \leq 12$$

$$2x_1 + 3x_2 \leq 12$$

$$2x_1 - x_2 \geq -2$$

$$x_1 \leq 4$$

whereas $x_1, x_2 \geq 0$.

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Describe a method of solving mixed L. P. P.

(b) Use Big-M method to solve the following function :

Minimize :

$$z = 12x_1 + 20x_2$$

(3)

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Subject to :

$$6x_1 + 8x_2 \geq 100$$

$$7x_1 + 12x_2 \geq 120$$

$$x_1, x_2 \geq 0$$

(c) Solve graphically :

Max. :

$$z = -5x_2$$

Subject :

$$x_1 + x_2 \leq 1$$

$$0.5x_1 - 5x_2 \leq -10$$

$$x_1, x_2 \geq 0$$

3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What are the characteristics and limitations of Operations Research ?
- (b) Find the initial basic feasible solution of the following Transportation problem by Vogel's approximation method :

Warehouse →

	W_1	W_2	W_3	W_4	Capacity
F_1	19	30	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18

Requirement

5 8 7 14

- (c) Explain the concept of degeneracy in Simplex method.

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) The Dream-Drink Company has to work out a minimum cost transportation schedule to distribute Creates of drinks from three of its factories X, Y and Z to its three warehouses A, B and C. The required particulars are given below. Find the least cost transportation schedule :

		From			Crates Available
		A	B	C	
To	X	75	50	50	1040
	Y	50	25	75	975
	Z	25	125	25	715
Crates Required		1300	910	520	2730

- (b) Use Big-M method to solve the following problem :

Minimize :

$$z = 12x_1 + 20x_2$$

Subject to :

$$6x_1 + 8x_2 \geq 100$$

$$7x_1 + 12x_2 \geq 120$$

$$x_1, x_2 \geq 0$$

- (c) What do you mean by feasible and infeasible and bounded and unbounded solutions ?
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Discuss the objective of Operations Research.
- (b) Find the dual of the following problem :
Minimise :

$$z = 40x_1 + 200x_2$$

Subject to :

$$40x_1 + 40x_2 \geq 60$$

$$3x_1 + 10x_2 \geq 60$$

$$8x_1 + 10x_2 \geq 80$$

- (c) Solve the following by NWCM :

		Warehouse			Supply,
		W_1	W_2	W_3	S_i
Plant	P_1	7	6	9	20
	P_2	5	7	3	28
	P_3	4	5	8	17
Demand, D_i		21	25	19	65